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Syllabus

Where's my syllabus to guide me through life?

Megan McCafferty

Syllabus

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- What is a syllabus?
- Course syllabus
- Detailed course syllabus

Warning!

- The following slides simply contain the syllabus of the course "as is"
- **Some topics are better detailed in other slides**

What is a syllabus?

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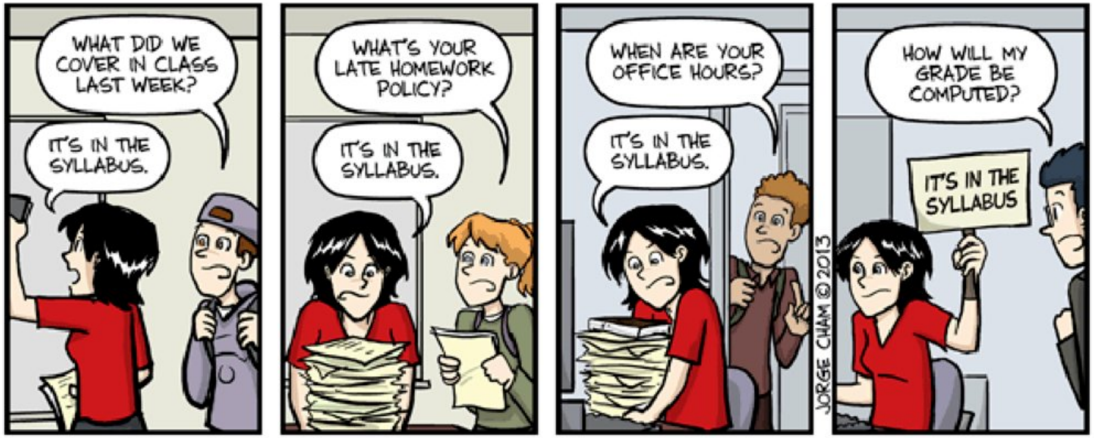
- A syllabus is a document that outlines all the essential information about a college course.
  - It lists the topics you will study
  - Lessons, tests, and exam modalities
  - Suggested books



What is a syllabus?

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- At the end of teaching activities you MUST fill in the survey about the course
- One question is about the syllabus
- Therefore, match the content of this presentation against what we have done in the lectures

Course content

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1. Introduction to programming
2. Algorithms
3. Basic computer architecture
4. Programming languages
5. The C programming language
6. Number representation in computer science
7. Expressions in C
8. Bitwise operators
9. Flow control
10. Data types in C
11. Predefined functions in C
12. Arrays
13. Dynamic memory allocation and pointers
14. Pointers to pointers
15. C strings
16. Functions
17. Computational complexity, search and sorting
18. I/O in C
19. Composite data types
20. Function pointers
21. Vibe coding and AI-assisted programming

Suggested Textbooks

- K.N. King, C Programming: A Modern Approach (2nd Edition), W W Norton & Co
- Bellini Guidi, Linguaggio C, Mc Graw Hill
- B.W. Kernighan e D. Ritchie, The C Programming Language: ANSI C Version, Pearson College
- Darnell Margolis, C manuale di programmazione, Mc Graw Hill

Education goals

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- The objective of the course is to provide the student with the ability to understand the principles of computer science and programming using C as the reference language and in particular:
  - Data representation
  - Algorithm concept
  - Basic architecture of processing systems
  - Procedural programming paradigm
  - Introduction to software engineering
- The skills to apply the knowledge listed concern the development of the so-called "computational thinking":
  - Decomposition of complex problems
  - Top-Down problem solving
  - Syntax and semantics of the C language

Prerequisites

- No propaedeutics courses
- It is assumed that the student knows the basics of using computers and the Internet
  - the equivalent of modules 1, 2, 3 and 7 of the ECDL (European Computer Driving Licence) syllabus.

Teaching methods

- Classroom lessons, with the help of slides made available to students in advance.
- Guided solution of classroom exercises.
- Programming exercises in the laboratory.
- Laboratory exercises are the most critical part of the course.
  - The proposed exercises are relayed to the same general topics as the classroom lessons.
  - The idea is to introduce the principles of programming, guiding the student to the solution of problems with an increasing level of complexity.

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